ChatGPT

Persistence Schema Contract

This document describes the recommended database schema for the RAGenius Execution Subsystem. The goal is to store execution state, skill definitions, workflow steps, tool calls, permission policies, MCP providers, and logs in a relational database (e.g. PostgreSQL) in a way that supports querying, auditing, recovery and scale.

Design Principles

- **Normalized structure:** Separate entities into distinct tables (Executions, Steps, Tool Calls, Skills, Tools, Policies, Providers, Logs). This avoids duplication and ensures referential integrity.
- **JSON fields for dynamic schemas:** Use JSON or JSONB columns for skill input/output schemas, workflow definitions and tool manifests, enabling flexible evolution without schema changes.
- **Timestamps and lifecycle:** Record creation and update timestamps for auditing and event ordering.
- **Indexes:** Create indexes on frequently queried fields (e.g. status, execution ID, skill ID, created_at) to ensure performance.
- **Foreign keys and constraints:** Enforce relationships between executions, steps and tool calls, and ensure referential integrity when deleting or updating records.

Entity Tables

`executions`

Column	Type	Constraints	Description
`id`	UUID	Primary key	Execution identifier.
`app_id`	text	Not null	Application that initiated the execution.
`session_id`	text	Not null	Session associated with the execution.
`skill_id`	text	Not null	References `skills.id`.
`status`	text	Not null, check in allowed values	Current status (`queued`, `running`, `completed`, `failed`, `partial`, `blocked`, `pending_confirmation`).
`result_type`	text	Nullable	Type of result (`text`, `json`, `file`, `video`).
`result`	jsonb	Nullable	Normalized result object.
`files`	jsonb	Nullable	List of file references.
`errors`	jsonb	Nullable	List of error objects.

Column	Type	Constraints	Description
`logs_summary`	text	Nullable	High-level summary of execution steps.
`created_at`	timestamptz	Not null, default `now()`	When the execution was created.
`started_at`	timestamptz	Nullable	When execution started.
`completed_at`	timestamptz	Nullable	When execution completed.
`updated_at`	timestamptz	Not null, default `now()`	When the record was last updated.

Indexes:

- `idx_executions_app_id` on (`app_id`)
- `idx_executions_session_id` on (`session_id`)
- `idx_executions_skill_id` on (`skill_id`)
- `idx_executions_status` on (`status`)
- `idx_executions_created_at` on (`created_at`)

`workflow_steps`

Column	Type	Constraints	Description
`id`	UUID	Primary key	Unique step identifier.
`execution_id`	UUID	Not null, references `executions.id`	Execution to which this step belongs.
`step_id`	text	Not null	Step identifier from workflow definition.
`step_type`	text	Not null	`validation`, `tool_call`, `local_decision`, `internal_workflow`, `service_call`, `saga`, or `end`.
`status`	text	Not null	`pending`, `running`, `completed`, `failed`, `skipped`.
`input_summary`	jsonb	Nullable	Summarized input payload for the step.
`output_summary`	jsonb	Nullable	Summarized output payload.
`error`	jsonb	Nullable	Error details if the step failed.
`started_at`	timestamptz	Nullable	Step start time.
`completed_at`	timestamptz	Nullable	Step end time.

Column	Type	Constraints	Description
`created_at`	timestamptz	Not null, default `now()`	When the record was created.

Index: `idx_workflow_steps_execution_id` on (`execution_id`).

`tool_calls`

Column	Type	Constraints	Description
`id`	UUID	Primary key	Unique call identifier.
`execution_id`	UUID	Not null, references `executions.id`	Execution to which this call belongs.
`step_id`	text	Not null	Step ID that invoked this tool.
`tool_id`	text	Not null, references `tools.id`	Identifier of the tool.
`provider_type`	text	Not null	Provider type of the tool.
`status`	text	Not null	`started`, `completed`, `failed`, `timeout`, `blocked`.
`input_summary`	jsonb	Not null	Summary of the input passed to the tool.
`output_summary`	jsonb	Nullable	Summary of the output returned by the tool.
`error`	jsonb	Nullable	Error object if the call failed.
`duration_ms`	integer	Nullable	Execution time in milliseconds.
`created_at`	timestamptz	Not null, default `now()`	When the record was created.

Index: `idx_tool_calls_execution_id` on (`execution_id`).

`skills`

Column	Type	Constraints	Description
`id`	text	Primary key	Unique skill identifier.
`name`	text	Not null	Human-friendly name.
`version`	text	Not null	Semantic version.
`description`	text	Nullable	What the skill does.

Column	Type	Constraints	Description
`input_schema`	jsonb	Not null	JSON Schema describing the expected input.
`output_schema`	jsonb	Not null	JSON Schema describing the result.
`required_tools`	jsonb	Not null	List of tool identifiers.
`required_permissions`	jsonb	Not null	List of required permission scopes.
`workflow_definition`	jsonb	Not null	Declarative description of the workflow steps.
`enabled`	boolean	Not null, default `true`	Whether the skill is active.
`created_at`	timestamptz	Not null, default `now()`	Creation time.
`updated_at`	timestamptz	Not null, default `now()`	Last update time.

Index: `idx_skills_name_version` on (`name`, `version`).

`tools`

Column	Type	Constraints	Description
`id`	text	Primary key	Unique tool identifier.
`name`	text	Not null	Human-friendly name.
`provider_type`	text	Not null	`local`, `api`, `mcp`, `rag_adapter`.
`input_schema`	jsonb	Not null	JSON Schema of expected inputs.
`output_schema`	jsonb	Not null	JSON Schema of outputs.
`permission_scopes`	jsonb	Not null	List of scopes required to call the tool.
`timeout_ms`	integer	Nullable	Timeout in milliseconds.
`side_effecting`	boolean	Not null	Whether the tool has side effects.
`metadata`	jsonb	Nullable	Provider-specific metadata.
`enabled`	boolean	Not null, default `true`	Whether the tool can be called.

Column	Type	Constraints	Description
`created_at`	timestamptz	Not null, default `now()`	Creation time.
`updated_at`	timestamptz	Not null, default `now()`	Last update time.

`permission_policies`

Column	Type	Constraints	Description
`id`	UUID	Primary key	Policy identifier.
`app_id`	text	Not null	Application to which this policy applies.
`tool_id`	text	Not null	Identifier of the tool.
`scope`	text	Not null	Permission scope.
`policy`	text	Not null	`auto_allow`, `restricted`, `require_confirmation`, `blocked`.
`conditions`	jsonb	Nullable	Additional constraints (see examples in auth document).
`created_at`	timestamptz	Not null, default `now()`	Creation time.
`updated_at`	timestamptz	Not null, default `now()`	Last update time.

Unique index: `(app_id, tool_id, scope)` to avoid duplicate policy entries.

`mcp_providers`

Column	Type	Constraints	Description
`id`	UUID	Primary key	Provider identifier.
`name`	text	Not null	Name of the MCP provider (e.g. `Notion`, `Asana`).
`server_url`	text	Not null	Base URL of the MCP server.
`auth_type`	text	Not null	`none`, `bearer`, `oauth2`, `mcp_session`.
`auth_config`	jsonb	Nullable	Configuration needed for authentication (client ID, scopes, etc.).
`discovered_tools`	jsonb	Nullable	Cached list of discovered tools.

Column	Type	Constraints	Description
`last_discovered_at`	timestamptz	Nullable	Last time discovery ran.
`enabled`	boolean	Not null, default `true`	Whether this provider is active.
`created_at`	timestamptz	Not null, default `now()`	Creation time.
`updated_at`	timestamptz	Not null, default `now()`	Last update time.

`execution_logs`

Column	Type	Constraints	Description
`id`	UUID	Primary key	Log record identifier.
`execution_id`	UUID	Not null, references `executions.id`	Execution to which this log belongs.
`level`	text	Not null	`debug`, `info`, `warn`, `error`, `audit`.
`event_type`	text	Not null	Nature of the event (e.g. `tool.called`, `execution.started`).
`message`	text	Not null	Human-readable description.
`summary`	jsonb	Nullable	Structured summary of event (e.g. tool ID, duration).
`created_at`	timestamptz	Not null, default `now()`	When the log was created.

Index: `idx_execution_logs_execution_id` on (`execution_id`).

Schema Evolution

- Use migrations (e.g. via Prisma, Flyway or Liquibase) to evolve the schema.
- For backwards-compatible changes (adding columns, adding nullable fields), create new columns with defaults.
- For breaking changes (renaming columns, changing types), plan a migration strategy that copies data to new tables and cleans up old structures.
- Consider using event sourcing for extremely dynamic workflows, but for most cases the relational model suffices.

Data Retention and Privacy

- Define retention periods for executions, logs and tool call summaries based on regulatory requirements and operational needs (e.g. 90 days, 1 year).
- Support purging of old logs or anonymizing sensitive data.
- Avoid storing full tool inputs or outputs when they may contain user secrets or large payloads. Store only summaries and file references.
